

Conformal Aerial Geo-Localization Through Visual Place Recognition

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Abstract—Visual place recognition (VPR) can be used for aerial geo-localization (VG), by matching aerial images against a database of geo-referenced satellite images. VG can be a valuable replacement or complement for navigation within environments where the use of Global Navigation Satellite Systems (GNSS) is denied. While deep learning methods have significantly advanced VPR performance, most provide point estimates without statistical guarantees. We apply the Conformal Prediction (CP) framework to retrieval-based VPR, formulating distance-based and cumulative similarity-based nonconformity measures that generate retrieval sets guaranteed to contain the true location with a user-specified probability. While prior conformal retrieval work requires uncertainty-aware feature extractors, limiting applicability to specific model families, we propose model-agnostic formulations that operate directly on metric space, enabling the use of any pretrained backbone. We further propose a novel uncertainty measure for fixed-size retrieval sets, based on CP, enabling uncertainty-aware decision making under fixed computational budgets. Experiments on two cross-view aerial/satellite datasets validate coverage guarantees and compare the efficiency of standard CP, Adaptive Prediction Sets, and calibrated fixed- k retrieval.

I. INTRODUCTION

Visual geo-localization enables estimating camera pose in known environments using reference imagery. For aerial robotics, VG provides critical alternative navigation when GNSS signals are weak, degraded, unavailable, or deliberately compromised through jamming and spoofing.

In contested environments, adversaries now routinely employ GNSS denial as standard electronic warfare. In 2011, Iran captured a US RQ-170 Sentinel drone through GPS spoofing [1], [2]. Regional conflicts demonstrate widespread GNSS radio frequency interference affecting hundreds of aircraft daily and degrading precision munitions [3], [4], making the use of GNSS-independent localization systems very compelling.

Beyond accuracy, safety-critical navigation demands calibrated uncertainty quantification. An incorrect position estimate reported with high confidence risks mission failure or loss of assets. Understanding localization reliability enables appropriate risk mitigation: requesting operator intervention, switching navigation modes, or aborting unsafe maneuvers.

VG can be broadly divided into 3 categories [5]: visual place recognition [6], visual alignment (VA) [7] and visual odometry (VO) [8]. Some solutions integrate different categories of techniques within a single framework [5], [9]. VPR is often used as an initialization method, a way to address the wake-up robot problem [10], or to correct drift.

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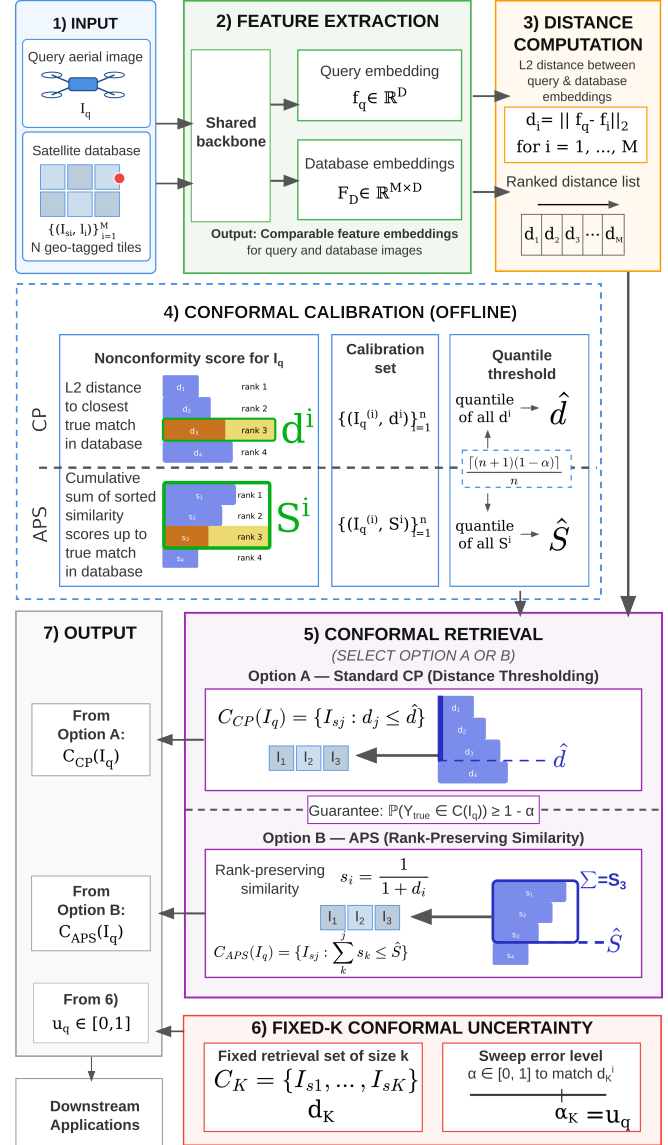


Fig. 1. Diagram of system architecture, from inputs (query and database images), to feature extraction, conformal calibration, distance computation for inference (with two variants, CP and APS) and also an uncertainty quantification branch for retrieval sets. The latter branch is not part of the standard CP inference pipeline, but an alternate use of the conformal framework for uncertainty quantification.

In this work, we use a content-based image retrieval pipeline to implement VPR, and apply it to aerial VG, with the goal of providing risk-controlled set estimates. We match a query aerial image from a downward-facing camera to a database of satellite images. The final estimator combines

the feature extraction model with the conformal prediction framework to provide coverage guarantees for the retrieval sets. Such guarantees could be useful when combined with other techniques such as multiple hypothesis tracking [11] or particle filtering [12], where the risk of not tracking of the correct hypotheses/particles might be controlled. While set sizes provide an intrinsic interpretation of instance uncertainty, we further explore the use of CP to provide uncertainty estimates for a fixed retrieval set size, applicable in scenarios where a fixed computational budget is imposed, e.g. real-time localization on embedded systems where a downstream task might refine the localization proposals.

While CP has been explored in the image retrieval problem [13], its applicability is constrained to uncertainty-aware feature extractors, limiting applicability to specific model families. Furthermore, while several uncertainty quantification techniques have been explored in the VPR space [14], using CP for estimating point-wise uncertainty from a fixed-size retrieval set is an underexplored area. Our contributions, illustrated in blocks 5 and 6 of Figure 1, aim to apply the CP framework to VPR for aerial VG, while simultaneously addressing the aforementioned limitations in previous work, and include:

- a model-agnostic conformal retrieval formulation for VPR that uses L2 distance as a nonconformity measure, departing from prior work [13] that requires uncertainty-aware feature extractors — this makes conformal VPR applicable to any pretrained backbone;
- an adaptation of Adaptive Prediction Sets (APS) from classification to distance-based retrieval through a rank-preserving similarity transform, yielding retrieval sets up to $12\times$ smaller than standard CP while maintaining coverage;
- a novel conformal uncertainty measure for fixed-size retrieval sets that provides calibrated per-query confidence estimates under bounded computational budgets;
- empirical validation on two aerial/satellite cross-view datasets (DenseUAV, AerialVL), demonstrating that coverage guarantees hold across varying calibration sizes, error rates, and matching radii.

A Background Section II briefly covers relevant topics concerning VPR, before tackling uncertainty estimation in the context of VPR and conformal retrieval in the Related Work Section III. In Section IV, we describe two conformal formulations for image retrieval (IV-A and IV-B), and the conformal uncertainty measure (IV-C). The VPR implementation and the conducted experiments are detailed in V, before presenting and discussing their results in VI.

II. BACKGROUND ON VPR

VPR is the task of recognizing a place from a given image. He et al. [5] identify VPR as one of three categories for VG. Similarly, Couturier et al. [15] place VPR as a subcategory in the large-scale matching category. It is usually posed as a retrieval problem, where a query image is matched to a database of images, but other approaches exist. This task can be challenging due to viewpoint changes, appearance changes,

and perceptual-aliasing [16], which can make it difficult to match images accurately.

Many works in VPR focus on matching a street-level query image to a database of street-level images [17]. Others have tried to match street-level queries to overhead map-like images [18], e.g. aerial or satellite images. In the context of aerial VG, the task is to match an aerial query image to a database of reference images, e.g. past aerial images, satellite images, maps.

VPR has been tackled in many forms, such as using hand-crafted features, e.g. SIFT [19]. Learning based approaches make up the bulk of recent literature on this topic, with many works that use deep learning to learn representations that tackle the challenges aforementioned [15]. Ahn et al. [20] use a siamese network to learn a similarity measure between different images of the same location. Representation learning is a common approach, where a model is trained to learn a representation that can be used to compare images with a simple L2 distance. Examples include the use of contrastive loss [21], [22], triplet loss [23] or its variants, e.g. soft triplet loss [24]. Recently, vision foundation models have been explored with promising results [25] even in zero-shot VPR.

VPR also benefits from the broader content-based image retrieval (CBIR) literature. CBIR is the task of analyzing the contents of query images and retrieving similar images from a database. While visual localization works have improved performance with specific techniques that handle the localization task better than generic CBIR approaches, there are some gaps in the literature which have been addressed earlier in CBIR, e.g. conformal retrieval, as will be detailed in Section III.

Formally, VPR can be defined as follows. A metric learning model f outputs a feature vector for both the query aerial image I_q and the satellite image I_s . The model is trained to minimize a distance $d(f(I_q), f(I_s))$ between the feature vectors of matching queries and database items (satellite images), while maximizing the distance when there is no match. A lower distance translates to a higher likelihood of a correct match between the query and the candidate.

III. RELATED WORK

A. Uncertainty in VPR

Quantifying uncertainty in computer vision, particularly within the realm of deep learning, has become an increasingly important area of research.

For VPR, Zaffar et al. [14] identify 4 categories of uncertainty estimation. In **retrieval based uncertainty**, the distance/similarity between the query and the first item of the retrieval set, or the ratio between the distances to the first and the second, are used. **Data driven** approaches include uncertainty-aware extractors. Examples that have been applied for retrieval include Bayesian Triplet loss [26] or STUN [27]. Other types of methods could also be appropriate if adapted for retrieval, such as variational autoencoders [28], where the variance in the latent space could be interpreted as uncertainty; Monte Carlo dropout [29], which approximates

Bayesian inference using dropout at test time; deep ensembles [30], using the variance of the outputs from multiple models; and evidential deep learning [31], which directly quantifies uncertainty by learning the parameters of a higher-order distribution, such as the Dirichlet distribution. **Geometric verification** can estimate uncertainty by using the number of inliers after a RANSAC loop using local descriptors (e.g. keypoints). **Spatial uncertainty** (SUE) [14] fits a Gaussian distribution over the poses of the retrieved set, weighted by their distances to the query. The variance of the fitted Gaussian serves as a proxy for the uncertainty.

SUE was proposed to address perceptual aliasing. In aerial VPR, the database will often have a location distribution close to uniform, i.e. roughly the same number of images per location. If the query is similar to multiple locations, the model will likely retrieve images from all those locations, which translates into clusters of retrieved locations, and not only a big cluster around the true location, which is not ideal and will make uncertainty estimates more uniform and less meaningful.

B. Conformal Retrieval

Predictors that are risk controlled give tight guarantees that a specified risk is controlled in some way. The CP framework offers the benefit of marginal coverage, with statistical guarantees. Instead of point predictions, e.g. a single class or value, it yields sets (for classification) or ranges (for regression) for a given significance level α , which are guaranteed to have the true value with probability $1 - \alpha$ [32].

CP has been successfully applied to several machine learning tasks, such as multilabel image classification, tumor segmentation, time-series weather prediction [33], and natural language processing [34]. It is model-agnostic and provides an inherent measure for uncertainty based on output set size. CP is an attractive method as it does not require any assumptions about the data distribution, the model's architecture or even the heuristic used to compute the nonconformity measure.

While conformal predictors are not probabilistic predictors, the size of prediction sets can be interpreted as a point estimate of uncertainty, which can be useful in many applications. Karimi et al. [35] use the set size to compute a measure of uncertainty. Its interpretation is straightforward and intuitive, depending on both α and the set size. However, in large-scale image retrieval, good representations typically yield small conformal sets relative to database size. In our experiments, retrieval set sizes vary considerably, but remain small relative to the total database. This results in estimated uncertainties that are all fairly small and similar, even when one retrieval set is twice the size of another.

Another approach computes p-values for predictions and transforms those into probabilities [36]. The authors showcase its application to label conditioned conformal prediction in a classification scenario. The label conditioned p-value for a test input x , to a given label $y \in \mathbf{Y}$ is given by the number of items in the calibration set that belong to that label y and whose conformity scores for that label α^y are below the

conformity score of the test input to that label, divided by the total number of items that belong to label y . This formulation is not adequate for retrieval-based VPR, as the number of locations (interpreted as labels) can be very large, and not all locations will be represented in the calibration set (in fact, most will not).

CP has also been successfully applied to retrieval within NLP contexts [37], [38]. Closer to the VPR context, Cai et al. [13] have applied CP to image retrieval, using it to control the risk of all true neighbors of a query image being outside the retrieved set. Their technique is based on uncertainty-aware feature extractors, which measure how confidently the computed features represent the images. The uncertainty estimate is used to model the size of the final retrieval set using a variant of Adaptive Prediction Sets.

C. Data

Datasets for aerial visual localization have recently increased in number and size, as unmanned aerial vehicles (UAVs) have become more popular and GNSS-less localization has become a hot topic. Most datasets are based on aerial images, captured from small commercial UAVs, and satellite images, aggregated from public sources. Datasets with high-altitude aerial images are scarce and coastal representation is missing altogether, as far as we were able to find. Couturier et al. [15] provide an overview of many existing relevant datasets (as well as the broader aerial visual localization field). We highlight the two datasets used in this work: DenseUAV [6], containing low-altitude urban imagery; and AerialVL [5], a larger dataset with both rural and urban imagery.

IV. METHODS

This section details the two conformal retrieval formulations for image retrieval, and a conformal uncertainty measure for sets with fixed size.

A. Conformal VPR

As mentioned in Section III-B, CP has been applied to image retrieval before [13], with an approach that adjusts retrieval set size according to feature extractor uncertainty. Our key design choice, instead, is to use the L2 distance from query to database items directly as the nonconformity measure. This formulation can be observed within the overall system diagram in option A of block 5 in Fig. 1. This is motivated by two observations: firstly, it makes the conformal framework model-agnostic, applicable to any pretrained backbone without retraining; secondly, Zaffar et al. [14] report that a simple L2 distance between query and reference descriptors is a better estimate of image-matching uncertainty than data-driven approaches.

In VPR, CP provides a statistical guarantee that a true query-database match is included in the retrieval set with a certain probability. Given a desired error rate α , the conformal retrieval set $C(I_q)$ for a query image I_q satisfies the following coverage guarantee:

$$P(Y_{\text{true}} \in C(I_q)) \geq 1 - \alpha$$

where:

- Y_{true} represents a true match of the query image I_q , i.e. a satellite image of the same location as the query.
- $C(I_q)$ is the retrieval set constructed using the conformal prediction method.
- α is the desired error rate, which specifies the maximum allowable probability of a true match not being included in the retrieval set.

This holds under the assumption that the data is exchangeable, meaning that the joint distribution of the data points does not change with their order. This assumption is reasonable for single query retrieval, as the database is fixed during calibration and inference, and calibration queries and test queries are sampled from the same kinds of environments (but not the exact same locations).

The application of CP requires a calibration set $\mathcal{C}$, a disjoint set of labeled data used to calibrate the prediction sets. From this set, we can compute the nonconformity thresholds that will be used to determine the retrieval sets during inference. The calibration set is composed of pairs $(I_q^{(i)}, d^{(i)})$, where:

- $I_q^{(i)}$ is the i -th query image.
- $d^{(i)}$ is the distance (nonconformity value) from the query image $I_q^{(i)}$ to its closest true match in the database.

The threshold distance $\hat{d}$ is the $\frac{\lceil (n+1)(1-\alpha) \rceil}{n}$ quantile of the nonconformity scores $\{d^{(1)}, d^{(2)}, \dots, d^{(n)}\}$ in the calibration set, where n is the calibration set size. During inference, we use the following procedure to generate the retrieval sets:

- 1) for a new query image I_q , compute the distances $\{d_1, d_2, \dots, d_m\}$ to all m database images;
- 2) sort the distances $\{d_1, d_2, \dots, d_m\}$ in ascending order;
- 3) the retrieval set is defined by $C(I_q) = \{I_{si} : d(I_q, I_{si}) \leq \hat{d}\}$

In practice, steps 2 and 3 can be efficiently implemented using efficient data structures (e.g. KD-Tree, Ball Tree), or using specialized vector search frameworks such as FAISS [39].

B. Adaptive Retrieval Sets

The adaptability of set size is a desirable property, and we can accomplish it by using Adaptive Prediction Sets, by transforming the existing distance metric into a similarity metric. This formulation can be observed within the overall system diagram in option B of block 5 in Fig. 1. Our similarity transform uses following formula (which preserves ranking):

$$\text{similarity}(I_q, I_s) = \frac{1}{1 + d(I_q, I_s)}$$

The calibration set $\mathcal{C}$ now consists of pairs $(I_q^{(i)}, S^{(i)})$, where $S^{(i)}$ is the cumulative similarity score up to and including the closest true match, after sorting the similarities in descending order. The similarity score threshold $\hat{S}$ is, again, the $\frac{\lceil (n+1)(1-\alpha) \rceil}{n}$ quantile of the cumulative similarity scores $\{S^{(1)}, S^{(2)}, \dots, S^{(m)}\}$ in the calibration set. During inference:

- 1) for a new query image I_q , compute the similarity scores $\{s_1, s_2, \dots, s_n\}$ for all database images;

- 2) sort the similarity scores in descending order;
- 3) initialize cumulative similarity $S = 0$;
- 4) iterate over the sorted similarity scores, adding each similarity score to S and including the corresponding image in the retrieval set $C(I_q)$ while $S \leq \hat{S}$.

C. Conformal uncertainty for fixed k

The previous methods produce retrieval sets with variable size. At inference time, the size of those sets might be too large for downstream steps, such as geometric verification or multiple hypothesis tracking. For real-time application in autonomous systems, it is often useful to fix a computational budget for a given task, e.g. to ensure execution in real time. In such a scenario, one could define a maximum set size of k that would be feasible to apply some post processing step. This section explores how we can leverage the conformal prediction framework to quantify the probability of a fixed size retrieval set not including any true match for the query. Such an estimate could be used to decide whether it would be worthwhile to perform a downstream task or not, based on some success estimate threshold.

A straightforward approach is to sweep the acceptable error rate α during calibration, and compute the associated thresholds for each $\alpha \in]0, 1[$. In practice, this interval is discretized. During inference, we use the distances $d_1 \dots d_k$ and find the smallest α for which the fixed sized set is a match. This formulation can be observed within the overall system diagram in block 6 of Fig. 1.

Formally, the uncertainty estimate u_q of a given query is given by

$$u_q = \min\{\alpha : |\{I_i : d_i < t_\alpha, i = 1 \dots n\}| \leq k\}$$

where I_i are database images, d_i are the L2 distances from query to each I_i , and t_α is the threshold associated with each discretized α . The same formulation can be used with APS, but we'd use the cumulative similarities S_i instead of d_i .

V. EXPERIMENTAL WORK

This section describes the VPR pipelines used for training the models used to apply the described conformal techniques. It also details the datasets used and the experimental design for testing the proposed techniques.

A. VPR

We used two existing frameworks to fit VPR models on the selected datasets: the one from DenseUAV [6] and the deep visual geo-localization benchmark [40] (for the AerialVL dataset). These are frameworks used by the authors of each of the used datasets. Both frameworks shared the same overall architecture: feature extractor followed by an aggregation head, and used a representation learning loss, such as contrastive loss or triplet loss.

For training with DenseUAV [6], several backbones (from ResNets [41] to Vision Transformers [42]) and aggregators (GeM [43] and NetVLAD [44]) were used. The loss function was either contrastive loss [21], online triplet loss [23] or soft triplet loss [24]. This framework also included a

classification (cross-entropy) and distributional loss (Kullback-Leibler divergence). Dai et al. [6] reason that a classification loss helps the model learn discriminative features, leading to better representations. They use a class per location, and locations have significant overlap (hence the “dense” in “DenseUAV”). They claim this improves results, which we have confirmed, but we make 2 remarks: firstly, it is not scalable to datasets with tens of thousands or hundreds of thousands of locations, as the number of images per class could be very small (perhaps a single database item) - DenseUAV is fairly small, which is probably why this strategy works well; secondly, the significant overlap could lead to contradictory signals during optimization, since images of different locations can contain a large area in common, but classification and distributional losses would push the representations of those images apart as they are effectively in different classes.

For training with AerialVL [5], a similar approach was taken, but fewer models were trained given the size of the dataset. Only ResNet backbones were used (different sizes: 50, 101, 152) and only the GeM aggregation head was used followed by a fully connected layer to reduce final dimension size (512, 1024 or 2048). Offline triplet loss was the only loss used here.

B. Datasets

To empirically validate the proposed approach, datasets that include both aerial and satellite images of the same locations were used. We used two datasets: DenseUAV [6] and AerialVL [5].

The DenseUAV dataset represents mostly urban environments. There are 3033 locations in total, each with 3 query images (altitude of 80, 90 and 100 meters) and 2 satellite images (old and new) for each aerial image. The original test set contains 777 of these locations, and it was further split in half for calibration and testing. Aerial image height varies between 80 and 200 m, but we used only the images of a height of 80 m, since different heights differed very little. The AerialVL dataset collected data for both VPR and visual odometry. In total, there are 18,361 aerial images and 14,096 satellite images, taken from 11 flight sequences, half of which took trajectories of more than 5 km. Aerial image height varies between 120 and 200 m. The authors also make available an 11,000 image set for training, from a different source. Following the authors’ procedure, we trained on this set, but calibrated and tested on the main AerialVL set.

Each dataset is divided into three disjoint subsets: training, calibration, and testing. The training set is used to train the base VPR model, such as the embedding network responsible for extracting features. The calibration set is a hold-out set, completely separate from the training data, and its purpose is to calibrate the CP and APS methods.

C. Experiments

We conducted a series of experiments to evaluate the performance of CP and APS in terms of coverage guarantees and prediction set size.

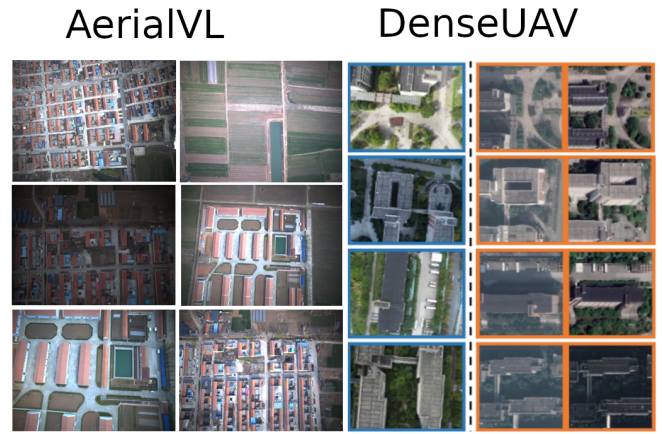


Fig. 2. Samples from the datasets AerialVL [5] and DenseUAV [6]

Because of significant overlap between images (datasets are gathered from flight trajectories), care was taken to ensure that no images from the same location were present in both the calibration and test sets. To this end, we visually inspected the overlap between queries at different distance thresholds, and selected a threshold that ensured no overlap. Afterwards, we compiled all independent locations. Each independent location was sampled by taking a random query and selecting all queries within the threshold distance of that location. Selected queries are removed from the pool of available queries for sampling. Several trials of this process were done, and we picked the one with the most images. The dataset size was reduced in this way, but no data leakage between calibration and test sets existed.

We employed k-fold cross validation. Each fold includes one or more independent locations. Locations were assigned to folds to ensure that no fold was smaller than 60% of the largest fold. Because folds have different sizes, results are a weighted average.

One of the effects being studied is the impact of the calibration set size. When all remaining folds are not used for calibration, the sample of the calibration set can impact significantly the calibration procedure. For that reason, we did 50 trials for each cross-validation run with a different sample of queries for the calibration set.

The experiments were conducted for different values of α (0.01, 0.05, 0.1), which specifies the maximum allowable probability of a true match not being included in the retrieval set within the CP framework. We also varied the radius for considering a match as correct, using values of 10, 50, 100 meters, which significantly impacts recall.

CP and APS were compared with another nonconformal method, mentioned as calibrated fixed- k . The $1 - \alpha$ quantile of the rank of the closest true match of each query in the calibration set was computed. During inference, the top- k database items are retrieved, regardless of the distance from query to the database.

VI. RESULTS

The results show trends over all trials and are presented for the test set of each dataset.

A. Conformal retrieval

Both CP and APS were applied to the test set, and the results were compared in terms of coverage and retrieval set size.

Coverage guarantees On average, both CP and APS correctly provided coverage guarantees in most cases, i.e. the fraction of retrieval sets that contain a true match to the query is above $1 - \alpha$ (for each trial). However, the error rate was not always below α for all trials, as can be seen in Fig. 3 by the curves that have mass to the right of the α threshold, but was never significantly above it either. The same figure shows how the error rate is distributed across different sizes of the calibration set. The error rate is lower for the smallest calibration set (in the conformal methods), because CP compensates by increasing the size of the retrieval sets, which is why the curve in the set size plots below is significantly wider.

As calibration size reaches a certain threshold (around 1000 samples), the distributions become very similar for both error rate and set size. This suggests that increasing calibration set sizes above a certain point does not yield further improvements in these metrics. The simple calibrated fixed- k method enjoys the least difference regarding the distribution of set sizes. During its calibration, the used quantile is the same regardless of calibration size. However, the conformal methods dynamically adjust the quantile according to the size of the provided calibration set. A smaller calibration set yields more conservative quantiles, which makes it harder to err during inference by significantly increasing retrieval set size. This is made even more obvious by the correlation between error rate and calibration size, after controlling for α and the recall of the base model. Conformal methods have a positive correlation (between 0.35 and 0.45), while calibrated fixed- k has no correlation at all.

Prediction Set Size Beyond coverage guarantees, retrieval set size is an important metric, as smaller sets are generally more useful in practice, as they reduce the computational burden on downstream tasks that may process all images in the retrieval set. The difference between set sizes for each model was highest in the AerialVL dataset. On average, sizes for CP were 15.2 and 11.9 times bigger than fixed- k and APS, respectively, while APS was 1.8 times bigger than fixed- k . On DenseUAV, sizes for CP were 3.6 and 2.2 times bigger than fixed- k and APS, respectively, while APS was 1.7 times bigger than fixed- k . This is not surprising, as standard split CP is not optimized to produce smaller prediction sets, which means the advantage is just the statistical guarantees. As the calibration size increases, the difference between conformal methods and fixed- k diminishes (correlation between -0.2 and -0.4). Models with high recall also produce smaller sets, which is not surprising. Although CP/APS guarantee coverage, the price to pay is larger sets, especially with low recall models.

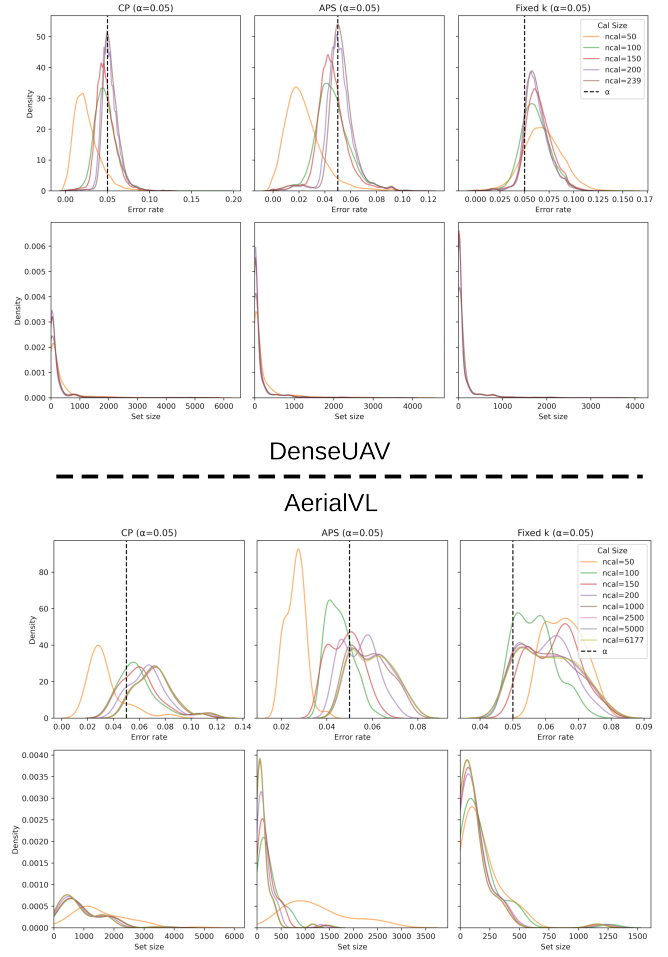


Fig. 3. Distribution (across all trials) of the error rate and set size for different methods and calibration sizes, with $\alpha = 0.05$. Computed with kernel density estimation.

Empty Sets are relevant because both CP and APS produce sets of variable length, and that length may be zero. The length of retrieval sets may be interpreted as a measure of uncertainty, with bigger sets indicating more uncertainty. A full set means the model retrieves all items to meet the desired error rate. On the other hand, an empty set means no retrieval can be made for the specified α . CP produced few empty sets, with a mean across all trials of 4% and 1% for AerialVL and DenseUAV, respectively. APS, on the other hand, produced no empty sets. Empty set frequency is positively correlated with recall performance.

Overall, the use of APS is preferred to that of CP. The set sizes that standard CP yields are not compatible with an application with real-time requirements. Still, if a given application does not require tight theoretical guarantees, using a simple approach such as calibrated fixed- k may be preferable, as the sizes of the retrieval sets are more manageable and performance is almost on par with the remaining methods.

B. Conformal uncertainty measure

When fixing the size of the retrieval set, our experiments suggest that the proposed method is more appropriate than other conformal uncertainty estimates, such as the technique proposed by Karimi et al. [35], as their estimates don't vary significantly for large retrieval databases. We fixed the set size to 20, 50, 100 and 200. Our method is illustrated in Fig. 4. The Expected Calibration Error (ECE) [45], a heuristic for how much the predicted confidence deviates from actual performance, was used to compare different trials. The ECE decreases as the size of the retrieval set increases, going from an average of 0.19 to 0.06. Furthermore, the ECE has significant negative correlation with the calibration size (-0.47), which is expected as a better calibration leads to lower error. The higher ECE in smaller retrieval sets could be addressed with further confidence calibration, e.g. histogram binning [46].

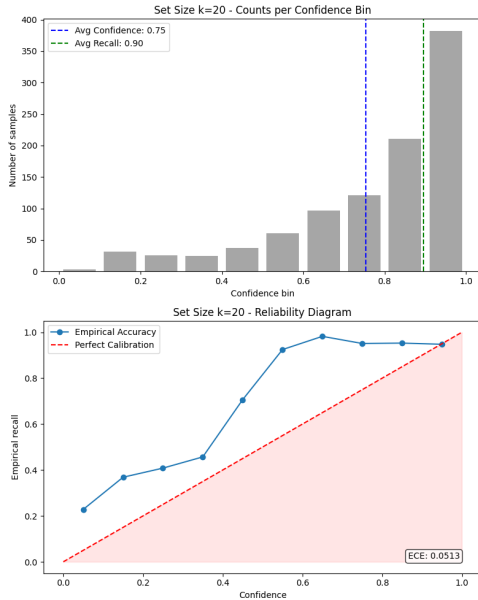


Fig. 4. Reliability diagram of a trial for $k = 20$. Above is the histogram of samples across confidence bins. Dataset is AerialVL.

VII. CONCLUSION

In this work, we investigated the application of conformal prediction to aerial visual place recognition for visual self geo-localization. We proposed two formulations—standard split CP with L2 distance-based nonconformity and Adaptive Prediction Sets with similarity-based accumulation, and evaluated them on the DenseUAV and AerialVL datasets. We also proposed a new conformal uncertainty measure for fixed sets, which may be useful in real-time applications with a bounded budget for processing time or amount of data to post-process.

Our experiments showed that both methods provide meaningful risk control: empirical coverage generally matched the desired confidence level, validating the statistical guarantees of CP in this setting. APS, in particular, produced much

smaller retrieval sets than standard CP, highlighting its potential for more efficient retrieval. We also observed that retrieval set size can be naturally interpreted as an instance-wise measure of uncertainty, although its expressiveness is limited when database sizes are very large. The simpler calibrated fixed- k method yielded similar error rates, while keeping the retrieval sets smaller, but lacks the strong theoretical guarantees of the conformal prediction framework. Lastly, the proposed uncertainty measure was shown to have a fairly low expected calibration error, especially for larger retrieval sets.

Follow-up work includes exploring other conformal approaches (e.g. Regularized APS), studying how CP can be used with other uncertainty quantification techniques (e.g. spatial or evidential), and applying it within downstream tasks, such as uncertainty-aware path planning and localization filters.

REFERENCES

- [1] S. Peterson and P. Faramarzi, "Iran hijacked US drone, says Iranian engineer," *Christian Science Monitor*, Dec. 2011. [Online]. Available: <https://www.csmonitor.com/World/Middle-East/2011/1215/Exclusive-Iran-hijacked-US-drone-says-Iranian-engineer>
- [2] D. Shepard, J. Bhatti, and T. Humphreys, "Drone hack," *GPS World*, vol. 23, no. 8, pp. 30–33, 2012. [Online]. Available: <https://www.gpsworld.com/drone-hack/>
- [3] S. C. Lo, Z. Liu, L. Ibrahim, Y. H. Chen, and T. Walter, "Observations of GNSS Spoofing in Russia in 2023-2024," *The International Technical Meeting of the The Institute of Navigation*, 2025. [Online]. Available: <https://api.semanticscholar.org/CorpusID:276389876>
- [4] S. Harris, "Russian jamming of U.S. weapons in Ukraine forces Pentagon to adjust," *The Washington Post*, May 2024. [Online]. Available: <https://www.washingtonpost.com/world/2024/05/24/russia-jamming-us-weapons-ukraine/>
- [5] M. He, C. Chen, J. Liu, C. Li, X. Lyu, G. Huang, and Z. Meng, "AerialVL: A Dataset, Baseline and Algorithm Framework for Aerial-Based Visual Localization With Reference Map," *IEEE Robotics and Automation Letters*, vol. 9, no. 10, pp. 8210–8217, Oct. 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/10632587/>
- [6] M. Dai, E. Zheng, Z. Feng, L. Qi, J. Zhuang, and W. Yang, "Vision-based UAV self-positioning in low-altitude urban environments," vol. 33, pp. 493–508, conference Name: IEEE Transactions on Image Processing. [Online]. Available: <https://ieeexplore.ieee.org/document/10376356>
- [7] H. Goforth and S. Lucey, "GPS-denied UAV localization using pre-existing satellite imagery," in *2019 International Conference on Robotics and Automation (ICRA)*, pp. 2974–2980, ISSN: 2577-087X. [Online]. Available: <https://ieeexplore.ieee.org/document/8793558>
- [8] C. Campos, R. Elvira, J. J. G. Rodriguez, J. M. M. Montiel, and J. D. Tardos, "ORB-SLAM3: An accurate open-source library for visual, visual-inertial, and multimap SLAM," vol. 37, no. 6, pp. 1874–1890. [Online]. Available: <https://ieeexplore.ieee.org/document/9440682/>
- [9] W. Xu, D. Yang, J. Liu, Y. Li, and M. Zhou, "A visual navigation algorithm for UAV based on visual-geography optimization," vol. 8, no. 7, p. 313. [Online]. Available: <https://www.mdpi.com/2504-446X/8/7/313>
- [10] J. Kinnari, R. Renzulli, F. Verdoja, and V. Kyrki, "LSVL: Large-scale season-invariant visual localization for UAVs," vol. 168, p. 104497. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0921889023001367>
- [11] M. Piasecki, "Global localization for mobile robots by multiple hypothesis tracking," *Robotics and Autonomous Systems*, vol. 16, no. 1, pp. 93–104, 1995, intelligent Robotics Systems SIRS '94. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S092188909500037G>
- [12] A. Şanlı, H. T. Bağcı, and E. Koyuncu, "Particle filter-based localization using visual feature synchronization in gnss-denied navigation," in *2025 Integrated Communications, Navigation and Surveillance Conference (ICNS)*, April 2025, pp. 1–8.

- [13] K. Cai, C. X. Lu, X. Zhao, and X. Huang, "Risk Controlled Image Retrieval," Jan. 2025, arXiv:2307.07336 [cs]. [Online]. Available: <http://arxiv.org/abs/2307.07336>
- [14] M. Zaffar, L. Nan, and J. F. Kooij, "On the estimation of image-matching uncertainty in visual place recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 17 743–17 753.
- [15] A. Couturier and M. A. Akhloufi, "A Review on Deep Learning for UAV Absolute Visual Localization," *Drones*, vol. 8, no. 11, p. 622, Oct. 2024. [Online]. Available: <https://www.mdpi.com/2504-446X/8/11/622>
- [16] S. Lowry, N. Sünderhauf, P. Newman, J. J. Leonard, D. Cox, P. Corke, and M. J. Milford, "Visual place recognition: A survey," *IEEE Transactions on Robotics*, vol. 32, no. 1, pp. 1–19, 2015.
- [17] W. Maddern, G. Pascoe, C. Linegar, and P. Newman, "1 Year, 1000km: The Oxford RobotCar Dataset," *The International Journal of Robotics Research (IJRR)*, vol. 36, no. 1, pp. 3–15, 2017. [Online]. Available: <http://dx.doi.org/10.1177/0278364916679498>
- [18] Y. Shi and H. Li, "Beyond Cross-view Image Retrieval: Highly Accurate Vehicle Localization Using Satellite Image," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. New Orleans, LA, USA: IEEE, Jun. 2022, pp. 16 989–16 999. [Online]. Available: <https://ieeexplore.ieee.org/document/9880032/>
- [19] F. Aznar, M. Pujol, and R. Rizo, "Visual Navigation for UAV with Map References Using ConvNets," in *Advances in Artificial Intelligence*. Cham: Springer International Publishing, 2016, pp. 13–22.
- [20] S.-H. Ahn, H.-S. Kang, and J.-M. Lee, "Satellite Image-based UAV Localization using Siamese Neural Network," *Proceedings of International Conference on Artificial Life and Robotics*, vol. 26, pp. 513–516, Jan. 2021. [Online]. Available: <https://alife-robotics.co.jp/LP/2021/OS4-1.htm>
- [21] A. Dosovitskiy, J. T. Springenberg, M. Riedmiller, and T. Brox, "Discriminative unsupervised feature learning with convolutional neural networks," *Advances in neural information processing systems*, vol. 27, 2014.
- [22] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proceedings of the 37th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 119. PMLR, 13–18 Jul 2020, pp. 1597–1607. [Online]. Available: <https://proceedings.mlr.press/v119/chen20j.html>
- [23] F. Schroff, D. Kalenichenko, and J. Philbin, "Facenet: A unified embedding for face recognition and clustering," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 815–823.
- [24] Y. Yuan, W. Chen, Y. Yang, and Z. Wang, "In defense of the triplet loss again: Learning robust person re-identification with fast approximated triplet loss and label distillation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 2020, pp. 354–355.
- [25] N. Keetha, A. Mishra, J. Karhade, K. M. Jatavallabhula, S. Scherer, M. Krishna, and S. Garg, "AnyLoc: Towards Universal Visual Place Recognition," *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1286–1293, Feb. 2024. [Online]. Available: <https://ieeexplore.ieee.org/document/10361537>
- [26] F. Warburg, M. Jorgensen, J. Civera, and S. Hauberg, "Bayesian triplet loss: Uncertainty quantification in image retrieval," in *2021 IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, pp. 12 138–12 148. [Online]. Available: <https://ieeexplore.ieee.org/document/9710535/>
- [27] K. Cai, C. X. Lu, and X. Huang, "Stun: Self-teaching uncertainty estimation for place recognition," in *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022, pp. 6614–6621.
- [28] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," *CoRR*, vol. abs/1312.6114, 2013. [Online]. Available: <https://api.semanticscholar.org/CorpusID:216078090>
- [29] Y. Gal and Z. Ghahramani, "Dropout as a bayesian approximation: Representing model uncertainty in deep learning," in *International Conference on Machine Learning*, 2015. [Online]. Available: <https://api.semanticscholar.org/CorpusID:160705>
- [30] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Neural Information Processing Systems*, 2016. [Online]. Available: <https://api.semanticscholar.org/CorpusID:6294674>
- [31] Y. Qin, D. Peng, X. Peng, X. Wang, and P. Hu, "Deep evidential learning with noisy correspondence for cross-modal retrieval," *Proceedings of the 30th ACM International Conference on Multimedia*, 2022. [Online]. Available: <https://api.semanticscholar.org/CorpusID:252783044>
- [32] V. Vovk, A. Gammernan, and G. Shafer, *Algorithmic Learning in a Random World*. New York: Springer-Verlag, 2005. [Online]. Available: <http://link.springer.com/10.1007/b106715>
- [33] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal Risk Control," in *Proceedings of the The Twelfth International Conference on Learning Representations*, Oct. 2024. [Online]. Available: <https://openreview.net/forum?id=33XGfHLtZg>
- [34] M. Campos, A. Farinhas, C. Zerva, M. A. T. Figueiredo, and A. F. T. Martins, "Conformal prediction for natural language processing: A survey," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 1497–1516, 2024. [Online]. Available: <https://aclanthology.org/2024.tacl-1.82/>
- [35] H. Karimi and R. Samavi, "Quantifying deep learning model uncertainty in conformal prediction," vol. 1, no. 1, pp. 142–148. [Online]. Available: <http://arxiv.org/abs/2306.00876>
- [36] V. Vovk, I. Petej, and V. Fedorova, "From conformal to probabilistic prediction," in *Artificial Intelligence Applications and Innovations*, L. Iliadis, I. Maglogiannis, H. Papadopoulos, S. Sioutas, and C. Makris, Eds. Springer, pp. 221–230.
- [37] P. Rouzrokh, S. Faghani, C. U. Gamble, M. Shariatnia, and B. J. Erickson, "CONFLARE: CONFormal LARge language model REtrieval," Apr. 2024, arXiv:2404.04287 [cs]. [Online]. Available: <http://arxiv.org/abs/2404.04287>
- [38] S. Li, S. Park, I. Lee, and O. Bastani, "TRAC: Trustworthy Retrieval Augmented Chatbot," *Proceedings of the 40th International Conference on Machine Learning*, 2023.
- [39] J. Johnson, M. Douze, and H. Jégou, "Billion-scale similarity search with gpus," *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2019.
- [40] G. Berton, R. Mereu, G. Trivigno, C. Masone, G. Csurka, T. Sattler, and B. Caputo, "Deep visual geo-localization benchmark," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 5396–5407.
- [41] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.
- [42] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations (ICLR)*, 2021. [Online]. Available: <http://arxiv.org/abs/2010.11929>
- [43] F. Radenović, G. Tolias, and O. Chum, "Fine-tuning cnn image retrieval with no human annotation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 41, no. 7, pp. 1655–1668, July 2019.
- [44] R. Arandjelović, P. Gronat, A. Torii, T. Pajdla, and J. Sivic, "Netvlad: Cnn architecture for weakly supervised place recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 40, no. 6, pp. 1437–1451, June 2018.
- [45] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *International conference on machine learning*. PMLR, 2017, pp. 1321–1330.
- [46] C. Gupta and A. Ramdas, "Distribution-free calibration guarantees for histogram binning without sample splitting," in *International conference on machine learning*. PMLR, 2021, pp. 3942–3952.